

A Multi-Model Framework for Rainfall Forecasting: Evaluating Performance Model Statistical, Machine Learning, and Deep Learning Methods

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Abstract—Accurate rainfall prediction plays an important role in hydrometeorological disaster mitigation and water resource management. This study examines the most accurate model for predicting daily rainfall in Batam City, Indonesia. The present study evaluates statistical, machine learning, and deep learning models using BMKG data from 2019 to 2023. A sliding window technique is applied with six lag configurations (lag-2 to lag-7, denoted as P1–P6). The models include ARIMA, SARIMA, ETS, VAR, and MLR (statistical); RF, SVR, XGBoost (machine learning); MLP, FNN, LSTM, GRU, and TCN (deep learning). Performance is assessed using RMSE, MAE, and R^2 . Results show that deep learning models outperform statistical and machine learning approaches, with LSTM achieving the highest R^2 (0.12) at P6, demonstrating its strength in capturing long-term dependencies. GRU and TCN also perform well, while SVR achieves the lowest MAE (2.02–2.11), making it suitable for short-term forecasting. Statistical models perform poorly, with high RMSE (4.22–4.24) and negative R^2 values (-0.06 to -0.05), indicating limited predictive capability. Despite the relatively low R^2 values, the comparative framework provides valuable insights into model behavior and performance, serving as a reference for future rainfall forecasting studies, particularly in tropical regions with similar hydrometeorological characteristics. These findings suggest that LSTM and GRU are optimal for real-time rainfall forecasting, while hybrid models may further improve accuracy. Future studies should focus on hyperparameter optimization, transformer-based architectures, and integrating additional meteorological variables. This study supports the development of AI-driven early warning systems and enhanced climate resilience strategies.

Keywords—deep learning, machine learning, rainfall forecasting, statistical models, time series forecasting.

I. INTRODUCTION

Global climate change and increased human activities have significantly intensified the frequency of extreme weather events, including floods, droughts, and storms. Hydrometeorological disasters, particularly floods, have become a global concern due to their widespread impacts on social, economic, and environmental aspects [1]. As an archipelagic country with a tropical climate, Indonesia is

among the most vulnerable nations to hydrometeorological disasters, such as floods and landslides. Previous studies have reported rainfall variability plays a dominant role compared to temperature fluctuations, where extreme rainfall patterns often trigger hydrological disasters, including floods during the rainy season and prolonged droughts in the dry season, [2], [3], [4], [5].

One of the regions in Indonesia significantly affected by rainfall variability is Batam City. Over the past two decades, rainfall trends in this city have shown a consistent decline, particularly during peak rainfall months such as December and January. According to the Strategic Environmental Assessment (KLHS) for Batam City's Medium-Term Regional Development Plan (RPJMD) 2025–2030, this condition directly affects clean water availability, given that Batam heavily relies on reservoirs as its primary water source. Furthermore, despite decreasing rainfall, flood risks remain high due to suboptimal drainage systems and reduced water absorption areas caused by land conversion for industrial and residential purposes. These changes are also influenced by climatological factors, such as monsoon wind patterns, which introduce uncertainties in the spatial and temporal distribution of rainfall [1].

To address these challenges, developing an accurate rainfall prediction model is crucial. Various methods have been utilized to capture historical rainfall patterns and predict future changes, including statistical methods, machine learning, and deep learning approaches. Statistical models such as Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA) are employed to handle non-stationary data and seasonal patterns in rainfall [6], [7]. The Exponential Smoothing State Space Model (ETS) relies on exponential smoothing to capture trends and fluctuations, while Multiple Linear Regression (MLR) and Vector Autoregression (VAR) are used to analyze relationships between rainfall and other meteorological variables [8][9].

With advancements in artificial intelligence, machine learning models have been applied to enhance rainfall prediction accuracy. Random Forest (RF) and Support Vector Regression (SVR) are effective in handling complex nonlinear

relationships in meteorological data, whereas XGBoost employs boosting techniques to improve the performance of decision tree-based models [10], [11], [12], [13]. Additionally, deep learning models such as Multilayer Perceptron (MLP) and Feedforward Neural Network (FNN) have demonstrated their ability to capture intricate patterns in rainfall time series data [9], [14], [15]. Furthermore, advanced deep learning architectures, including Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Temporal Convolutional Network (TCN) have been proven effective in handling temporal dependencies and capturing long-term patterns in meteorological data [16], [17], [18].

This study aims to determine the most accurate model for predicting rainfall in Batam City by evaluating a range of statistical, machine learning, and deep learning approaches. Using a sliding window technique, historical rainfall time series data are transformed into a supervised learning format, where past observations serve as predictors for future rainfall values. The models evaluated in this study include ARIMA, SARIMA, ETS, VAR, and MLR from the statistical category; RF, SVR, XGBoost, MLP, and FNN from the machine learning category; and LSTM, GRU, and TCN from the deep learning category. The model performance is assessed using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2) to quantify prediction accuracy [19]. The findings of this study are expected to identify the most accurate model, which can serve as a basis for water resource management, flood mitigation, and the development of AI-based early warning systems in Batam City.

II. METHOD

This study adopts an experimental research approach to evaluate the accuracy of various rainfall prediction models in Batam City. A comparative analysis is conducted on three predictive model categories—statistical models, machine learning, and deep learning—to determine the most effective approach in modeling rainfall patterns using a dataset obtained from the Meteorology, Climatology, and Geophysics Agency (BMKG) Batam. The evaluation framework applied in this study is illustrated in Fig. 1, which presents the models and evaluation process used.

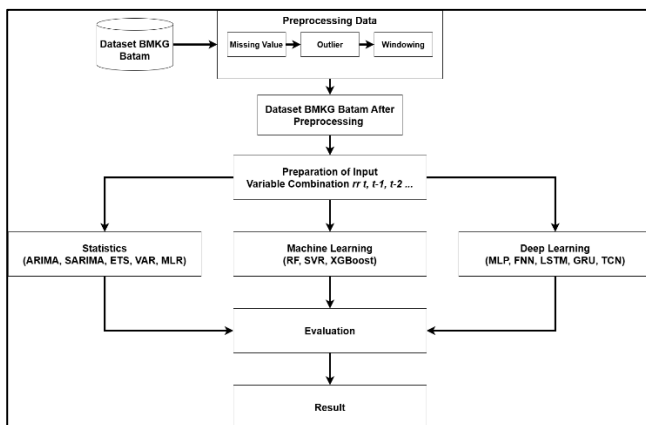


Fig. 1. Flow Chart Evaluating Performance Model Statistical, Machine Learning, and Deep Learning Methods

A. Dataset Structure

The dataset used in this study was acquired from BMKG Batam via the BMKG Data Online platform, containing daily

rainfall records from January 1, 2019, to December 31, 2023. Rainfall data serve as the primary variable in this study, as they play a crucial role in determining daily meteorological conditions and in constructing time series-based prediction models.

The dataset consists of:

- Date (observation date)
- Rainfall (rr) values (in mm)

The value of $rr = 0$ mm indicates no rainfall on that day, whereas $rr > 0$ mm represents the presence of rainfall with varying intensities. The dataset structure is presented in Table I.

TABLE I. RAINFALL DATASET FROM BMKG BATAM CITY

No	Date	rr
1	31/12/2023	1.6
2	30/12/2023	3.1
3	29/12/2023	23
4	28/12/2023	137.1
5	27/12/2023	3.6
...	...	...
1826	1/1/2019	0

B. Data Preprocessing

Data preprocessing is crucial to ensure the quality and consistency of the dataset before model training. The process includes handling missing values, outlier detection and mitigation, and data transformation using the sliding window technique to convert the time series into a supervised learning format.

1) Handling Missing Values

Missing values in time series data can impact model accuracy. In this study, missing values are handled using linear interpolation, which estimates missing values based on adjacent data points. Unlike mean or median imputation, linear interpolation preserves temporal trends, avoiding sudden distortions in rainfall patterns [20].

2) Outlier Detection and Mitigation

Outliers often result from extreme weather events or recording errors. These anomalies are detected using the Interquartile Range (IQR) method, and extreme values are replaced with the median. This approach prevents skewness and ensures robustness in model training [21].

3) Data Transformation Using the Sliding Window

To convert the time series dataset into a supervised learning format, this study employs the sliding window technique. This transformation enables the predictive models to learn temporal dependencies by using previous observations as input features for future rainfall predictions.

In this study, multiple input lag values are used to construct a new dataset with different past time steps as predictors, specifically:

- P1 (lag-2): Using rainfall values from two days prior.
- P2 (lag-3): Using rainfall values from three days prior.

- P3 (lag-4): Using rainfall values from four days prior.
- P4 (lag-5): Using rainfall values from five days prior.
- P5 (lag-6): Using rainfall values from six days prior.
- P6 (lag-7): Using rainfall values from seven days prior.

Formally, the dataset transformation follows the (1):

$$(x(t-7), x(t-6), x(t-5), x(t-4), x(t-3), x(t-2), x(t-1)) \rightarrow y(t) \quad (1)$$

where:

- $x(t-n)$ represents the rainfall measurement at time $t-n$,
- $y(t)$ is the target variable representing the rainfall at time t .

The choice of lag configurations (lag-2 to lag-7) was informed by exploratory autocorrelation (ACF) analysis [22], [23], which indicated that the rainfall data exhibited weak but non-negligible temporal dependencies within a 7-day window. These lag values balance model complexity and information depth, providing sufficient historical context without excessively increasing feature dimensionality.

By structuring the dataset in this way, the model can effectively learn short-term dependencies in the rainfall pattern. This approach ensures that historical rainfall trends are properly incorporated into the predictive models, enhancing their ability to forecast future rainfall values accurately.

Table II-VII illustrate the structured datasets created for different input lags (P1 to P6) using the sliding window technique.

TABLE II. DATASET FOR P21(LAG-2)

Date	$x(t-2)$	$x(t-1)$	$y(t)$
12/31/2023	1.2	3.1	1.6
12/30/2023	1.2	1.2	3.1
12/29/2023	3.6	1.2	1.2
12/28/2023	1.1	3.6	1.2
...	...	...	...
1/3/2019	1.2	4	0

TABLE III. DATASET FOR P2 (LAG-3)

Date	$x(t-3)$	$x(t-2)$	$x(t-1)$	$y(t)$
12/31/2023	1.2	1.2	3.1	1.6
12/30/2023	3.6	1.2	1.2	3.1
12/29/2023	1.1	3.6	1.2	1.2
12/28/2023	16.85	1.1	3.6	1.2
...	...	...	...	...
1/4/2019	1.2	4	0	4.3

TABLE IV. DATASET FOR P3 (LAG-4)

Date	$x(t-4)$	$x(t-3)$	$x(t-2)$	$x(t-1)$	$y(t)$
12/31/2023	3.6	1.2	1.2	3.1	1.6
12/30/2023	1.1	3.6	1.2	1.2	3.1
12/29/2023	16.85	1.1	3.6	1.2	1.2
12/28/2023	1.2	16.85	1.1	3.6	1.2
...	...	...	...	...	...

1/5/2019	1.2	4	0	4.3	0
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TABLE V. DATASET FOR P4 (LAG-5)

Date	$x(t-5)$	$x(t-4)$	$x(t-3)$	$x(t-2)$	$x(t-1)$	$y(t)$
12/31/2023	1.1	3.6	1.2	1.2	3.1	1.6
12/30/2023	16.85	1.1	3.6	1.2	1.2	3.1
12/29/2023	1.2	16.85	1.1	3.6	1.2	1.2
12/28/2023	0.1	1.2	16.85	1.1	3.6	1.2
...	...	...	...	...	...	...
1/6/2019	1.2	4	0	4.3	0	0

TABLE VI. DATASET FOR P5 (LAG-6)

Date	$x(t-6)$	$x(t-5)$	$x(t-4)$	$x(t-3)$	$x(t-2)$	$x(t-1)$	$y(t)$
12/31/2023	16.85	1.1	3.6	1.2	1.2	3.1	1.6
12/30/2023	1.2	16.85	1.1	3.6	1.2	1.2	3.1
12/29/2023	0.1	1.2	16.85	1.1	3.6	1.2	1.2
12/28/2023	1.6	0.1	1.2	16.85	1.1	3.6	1.2
...	...	...	...	...	...	...	...
1/7/2019	1.2	4	0	4.3	0	0	0

TABLE VII. DATASET FOR P6 (LAG-7)

Date	$x(t-7)$	$x(t-6)$	$x(t-5)$	$x(t-4)$	$x(t-3)$	$x(t-2)$	$x(t-1)$	$y(t)$
12/31/2023	1.2	16.85	1.1	3.6	1.2	1.2	3.1	1.6
12/30/2023	0.1	1.2	16.85	1.1	3.6	1.2	1.2	3.1
12/29/2023	1.6	0.1	1.2	16.85	1.1	3.6	1.2	1.2
12/28/2023	1.7	1.6	0.1	1.2	16.85	1.1	3.6	1.2
...	...	...	...	...	...	...	...	...
1/8/2019	1.2	4	0	4.3	0	0	0	4.2

C. Model Implementation

This study evaluates three categories of predictive models: statistical models, machine learning models, and deep learning models. Each model group has distinct advantages in capturing rainfall patterns.

1) Statistical Models

Statistical models capture linear trends and seasonal patterns in time series data. The following models are implemented: Autoregressive Integrated Moving Average (ARIMA), Seasonal ARIMA (SARIMA), Exponential Smoothing State Space Model (ETS), Vector Autoregression (VAR), and Multiple Linear Regression (MLR).

2) Machine Learning Models

Machine learning models are employed to capture non-linear dependencies in rainfall data. The models include Random Forest (RF), Support Vector Regression (SVR), Extreme Gradient Boosting (XGBoost).

3) Deep Learning Models

Deep learning models are utilized to capture long-term dependencies in time series data. The models include Multilayer Perceptron (MLP), Feedforward Neural Network (FNN), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Temporal Convolutional Network (TCN).

Each model is trained and evaluated using an 80:20 train-test split, where 80% of the data is used for training, and 20% is reserved for validation and testing.

D. Performance Evaluation

Model performance is assessed using three key metrics:

- Root Mean Square Error (RMSE): Measures the overall magnitude of prediction errors.
- Mean Absolute Error (MAE): Evaluates the average absolute error in predictions.
- Coefficient of Determination (R^2): Indicates how well the model explains rainfall variance.

The best-performing model is selected based on the lowest RMSE and MAE, and the highest R^2 value. Additionally, statistical significance tests are conducted to validate model robustness and generalization.

E. Result Reporting

This section presents a systematic evaluation of the models to provide a comprehensive understanding of the performance of each approach in rainfall prediction. The evaluation is conducted based on key metrics, including Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2), to assess the most effective model for rainfall forecasting.

III. RESULT AND DISCUSSION

This section presents the evaluation results of various rainfall prediction models across different lag inputs (P1 to P6) and discusses their effectiveness in terms of accuracy and reliability. The models are assessed using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2) to evaluate their forecasting capabilities. The performance of statistical, machine learning, and deep learning approaches is analyzed to provide insights into their suitability for rainfall prediction. While deep learning models demonstrated superior accuracy, they also required significantly greater computational resources and longer training times compared to statistical or machine learning models. This trade-off underscores the importance of considering both model complexity and computational cost when deploying predictive systems, particularly in resource-constrained environments. The complete evaluation results are presented in Table VIII.

TABLE VIII. EVALUATION OF RMSE, MAE, AND R^2 FOR VARIOUS RAINFALL PREDICTION MODELS USING LAG-BASED INPUTS

Model	P1			P2			P3			P4			P5			P6		
	RMSE	MAE	R2	RMSE	MAE	R2	RMSE	MAE	R2	RMSE	MAE	R2	RMSE	MAE	R2	RMSE	MAE	R2
ARIMA	4.23	3.37	-0.06	4.23	3.38	-0.06	4.24	3.38	-0.06	4.23	3.37	-0.06	4.23	3.37	-0.06	4.24	3.38	-0.06
SARIMA	4.22	3.35	-0.05	4.22	3.35	-0.06	4.23	3.36	-0.06	4.22	3.35	-0.06	4.22	3.35	-0.05	4.23	3.36	-0.06
ETS	4.15	2.52	-0.02	4.14	2.57	-0.01	4.13	2.55	-0.01	4.15	2.52	-0.02	4.15	2.52	-0.02	4.14	2.55	-0.01
VAR	4.24	3.39	-0.06	4.24	3.38	-0.06	4.24	3.38	-0.06	4.23	3.37	-0.06	4.23	3.35	-0.05	4.22	3.34	-0.05
MLR	4.04	3.12	0.03	4.03	3.10	0.04	4.02	3.07	0.04	4.00	3.04	0.05	4.01	3.01	0.05	3.99	2.99	0.06
RF	4.10	2.73	0.00	4.19	2.76	-0.04	4.11	2.71	0.00	4.15	2.81	-0.02	4.06	2.76	0.03	4.14	2.74	-0.01
SVR	4.06	2.02	0.02	4.06	2.04	0.02	4.07	2.06	0.02	4.09	2.06	0.01	4.11	2.07	0.00	4.13	2.11	-0.01
XGBoost	4.23	2.84	-0.06	4.34	2.82	-0.11	4.29	2.68	-0.09	4.26	2.80	-0.07	4.37	2.90	-0.13	4.32	2.85	-0.10
MLP	3.96	2.92	0.07	3.97	2.92	0.07	3.91	2.60	0.09	3.92	2.79	0.09	3.90	2.64	0.10	4.16	3.05	-0.02
FNN	3.91	2.77	0.10	3.91	2.52	0.10	4.03	2.73	0.04	3.98	2.79	0.06	4.01	2.70	0.05	4.44	2.82	-0.17
LSTM	3.96	2.97	0.07	3.97	2.92	0.07	3.94	2.86	0.08	3.94	2.92	0.08	3.89	2.71	0.11	3.85	2.69	0.12
GRU	3.92	2.85	0.09	3.89	2.61	0.10	3.99	3.05	0.06	3.91	2.81	0.10	3.90	2.75	0.10	3.95	2.90	0.08
TCN	3.95	2.92	0.08	3.98	3.02	0.06	3.91	2.81	0.10	3.90	2.84	0.10	3.92	2.85	0.09	3.96	2.95	0.08

A. Performance Analysis of Prediction Models

1) Statistical Models

Statistical models, including ARIMA, SARIMA, ETS, VAR, and MLR, generally exhibit lower predictive accuracy compared to machine learning and deep learning approaches.

- ARIMA and SARIMA have consistently high RMSE values (4.22 to 4.24) and negative R^2 scores (-0.06 to -0.05), indicating that they struggle to capture rainfall patterns effectively.
- ETS performs slightly better, achieving a lower RMSE (4.13 to 4.15) and less negative R^2 (-0.02 to -

0.01), suggesting its ability to capture some trends but still failing to generalize well.

- MLR shows slight improvements, particularly at P6 (RMSE: 3.99, R^2 : 0.06), demonstrating that simple linear regression models can leverage lagged features better than other statistical methods.

2) Machine Learning Models

Machine learning models, including SVR, RF, and XGBoost generally outperform statistical models, indicating their ability to learn non-linear relationships in the data.

- SVR achieves the lowest MAE (2.02 to 2.11) across all lag configurations, indicating its robustness in minimizing absolute errors. However, its RMSE (4.06 to 4.13) remains relatively high, suggesting some level of bias.
- XGBoost struggles with negative R^2 values, particularly at P5 (-0.13) and P6 (-0.10), which could indicate overfitting or instability in its predictions.

3) Deep Learning Models

Deep learning models, including MLP, FNN, LSTM, GRU, and TCN, provide the most consistent and accurate predictions across different lag configurations.

- FNN achieving an R^2 of 0.10 at P1 and P2, and MLP maintaining R^2 0.09 to 0.10 at P3-P5. However, MLP drops in performance at P6 (-0.02), possibly due to difficulty in capturing long-term dependencies.
- LSTM achieves the highest R^2 (0.12 at P6), with RMSE decreasing from 3.96 at P1 to 3.85 at P6, confirming its ability to model long-term dependencies in rainfall data.
- GRU performs similarly well, achieving an R^2 of 0.10 at P2-P5, but slightly declining at P6 (0.08).
- TCN remains competitive, maintaining an R^2 of 0.08 to 0.10 across all lag configurations.

Overall, LSTM outperforms all other models at P6, demonstrating the effectiveness of recurrent architectures in capturing sequential patterns in rainfall data.

B. Effect of Lag Input on Model Performance

The results indicate that the choice of lag configuration significantly impacts model performance.

- Statistical models show marginal improvements with increased lag values, but their overall performance remains weak.
- Machine learning models exhibit fluctuations, with SVR performing relatively consistently while XGBoost experiences significant performance degradation.
- Deep learning models benefit the most from extended lag inputs, particularly LSTM, which achieves its best performance at P6 ($R^2 = 0.12$).
- MLP and FNN show reduced effectiveness at longer lag values, suggesting that traditional feedforward networks may struggle with long-term dependencies.

These findings highlight the importance of selecting an appropriate lag configuration, as longer lags improve deep learning models while negatively impacting certain machine learning models.

C. Comparative Analysis of RMSE, MAE, and R^2

- RMSE: Deep learning models consistently achieve lower RMSE values, with LSTM at P6 (RMSE: 3.85) recording the lowest error. Machine learning models exhibit moderate improvements, while statistical models maintain the highest RMSE values.

- MAE: SVR consistently achieves the lowest MAE (~2.02-2.11), indicating its ability to minimize absolute errors, while deep learning models (LSTM, GRU, TCN) and neural network-based machine learning models (MLP, FNN) also demonstrate competitive performance.
- R^2 : The highest R^2 values are achieved by FNN (0.10 at P1), GRU (0.10 at P2-P5), and LSTM (0.12 at P6), confirming their ability to explain rainfall variability effectively.

D. Practical Implications and Applications

The findings suggest that deep learning models, particularly LSTM and GRU, are the most reliable for rainfall prediction. These models can be integrated into real-time forecasting systems to improve early warning mechanisms for extreme weather conditions.

Additionally, the results emphasize the importance of choosing optimal lag configurations, as longer lag inputs enhance deep learning models while affecting other approaches differently. Future research can explore:

- Hybrid models that integrate statistical and deep learning techniques for improved accuracy.
- Optimization of hyperparameters and architecture tuning to enhance deep learning models.
- Incorporation of additional meteorological variables (e.g., temperature, humidity) to improve prediction performance.

IV. CONCLUSION

This study evaluates the performance of statistical, machine learning, and deep learning models for rainfall prediction in Batam City, considering six different lag configurations (P1 to P6). The results indicate that deep learning models (LSTM, GRU, and TCN) consistently outperform statistical and machine learning approaches, particularly for longer lag inputs. LSTM achieves the highest R^2 (0.12) at P6, confirming its ability to model long-term dependencies in rainfall patterns, while GRU and TCN also demonstrate strong predictive capabilities. Among machine learning models, SVR achieves the lowest MAE (2.02 to 2.11), indicating its effectiveness in minimizing absolute errors, while MLP and FNN perform well in mid-range lag configurations but struggle with long-term dependencies. In contrast, statistical models (ARIMA, SARIMA, and ETS) exhibit high RMSE and negative R^2 values, indicating their limited ability to capture rainfall variability. The findings emphasize that deep learning models benefit from extended lag inputs, improving their accuracy in capturing sequential rainfall trends, whereas statistical models show only marginal improvements, and machine learning models exhibit mixed performance. The practical implications suggest that LSTM and GRU should be prioritized for real-time rainfall forecasting applications, while hybrid modeling approaches combining statistical and AI-based methods may further enhance accuracy. Future research should explore transformer-based models, hybrid techniques, and the integration of additional meteorological factors such as temperature and humidity. Overall, this study highlights the superiority of AI-driven approaches in hydrometeorological forecasting, paving the way for improved early warning

systems, water resource management, and disaster mitigation strategies in Batam City and similar regions worldwide.

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